

ProductDatabaseTest.java – Full Explanation (Tanglish + Colorful)

■ **File Name:** ProductDatabaseTest.java

■ **Overview:**

Ithu oru JUnit test class — database la iruka **Product** records ellam fetch panni test panna use pannurathu.

Purpose: Check panna, database connection, data retrieval and data integrity sariyaa iruka nu verify panna.

■ **@SpringBootTest:**

Ithu entire Spring context load pannum. So repository, service ellam automatic inject aagum.

■ **@Autowired:**

Spring autowire pannura — **ProductRepository** object automatically inject pannum test class la.

■ **Test Method: testReadAllProductsFromDatabase()**

■ Step-by-step explanation:

1■■■ `productRepository.findAll()` – Database la iruka ellaa product records fetch pannum.

2■■■ `assertNotNull(allProducts)` – Product list null aakudadhu nu check pannum.

3■■■ If list empty na, console la message print pannum: “No products found in DB.”

4■■■ Illa na, product count print pannum and foreach loop la product details print pannum.

5■■■ Each product la validate pannum:

■ Product name null illa

■ Model null illa

■ Price negative illa

■ Quantity negative illa

■ Status null illa

■ **Console Output Example:**

ID: 1, Name: iPhone, Model: 13 Pro, Price/Qty: 90000, Quantity: 10, Total: 900000, Status: ACTIVE

■ Purpose:

To confirm that DB connection, product table data and mappings ellam correct-a work pannuthu.

■ Summary:

- ✓ Ensures DB la data iruka nu verify pannum.
- ✓ Data fields integrity check pannum.
- ✓ Safe rollback mechanism illa – only read operation.